

DO Dinh Hieu

🌐 Homepage | @ dinhhieu.do.2020@smu.edu.sg | 🎓 Google Scholar | 🐙 GitHub

I am currently a Research Scientist at Preferred.AI where I work on user understanding and personalization for financial services. My expertise spans Recommender Systems, Machine Learning, Deep Learning, and Transformers. I am also working on agentic AI (LangGraph + Ollama) and RAG (LangChain + LCEL).

WORK EXPERIENCE

Preferred.AI | Research Scientist | November 2025 – Present | Singapore

- Designing systems and solutions for **user understanding and personalization** in financial services.
- Building **agentic AI** solutions with dozens of primitive tools for investor profiling using LangGraph and Ollama.
- Developing **RAG pipelines** for correctness-first data retrieval with LangChain and LCEL.

OSP JSC | AI Engineer Intern | May 2019 – April 2020 | Hanoi, Vietnam

- Developed various models, including **linear regression**, **ARIMA**, **XGBoost** to predict future user spending.
- Created a **real-time face recognition** system with **InsightFace embeddings** and a **cosine k-NN classifier**.
- Built a **Vietnamese news recommender system** powered by a **language model trained from scratch**.

EDUCATION

SMU School of Computing and Information Systems

PhD in Computer Science

Singapore

August 2020 – August 2025

Dissertation: Modeling Multiple Tasks in Recommendation Systems

VNU University of Engineering and Technology

BSc in Computer Science (Summa Cum Laude; Excellent Thesis Award)

Hanoi, Vietnam

August 2016 – July 2020

Thesis: Vietnamese News Recommendation Using Language Model

PUBLICATIONS

Compositions of Variant Experts for Integrating Short-Term and Long-Term Preferences

Jaime Hieu Do, Trung-Hoang Le, and Hady W. Lauw

ACM Transactions on Recommender Systems (TORS), To appear

Dual-Target Disjointed Cross-Domain Recommendation Mediated via Latent User Preferences

Jaime Hieu Do and Hady W. Lauw

ACM SIGKDD Explorations Newsletter, Volume 27, Issue 1, June 2025

Continual Collaborative Filtering Through Gradient Alignment

Jaime Hieu Do and Hady W. Lauw

Proceedings of the 17th ACM Conference on Recommender Systems (RecSys'23), September 2023

PROJECTS

Cornac | [GitHub](#) ~1,000 stars | Contributor | **Recommended by ACM RecSys 2023**

- Developed and optimized **PyTorch backend** for classical recommendation models (MF, FM, NCF), achieving competitive performance with Cython/C++ implementations, enhancing usability for advanced customization.
- Designed and implemented an **end-to-end pipeline** for **sequential recommendation**, encompassing data processing, model training, and evaluation with Transformer-based baselines like BERT4Rec and GPT4Rec.

AWARDS & ACHIEVEMENTS

PhD Full Scholarship from Ministry of Education, Singapore, 2020 – 2025

Excellent Thesis Award VNU University of Engineering and Technology, 2020

#7 in Hate Speech Detection Competition: Vietnamese Language and Speech Processing 2019

Full Scholarship for top 5% excellent students, University of Engineering and Technology, 2016 – 2020

ACADEMIC SERVICES & SKILLS

Reviewer: TORS

Programming: Python, PyTorch, LangGraph, LangChain, Ollama

Skills: Machine Learning, Recommender Systems, Deep Learning, Transformers, Agentic AI, RAG